Experiment 2(A): Transmission-Line / Two-Port S-Parameters

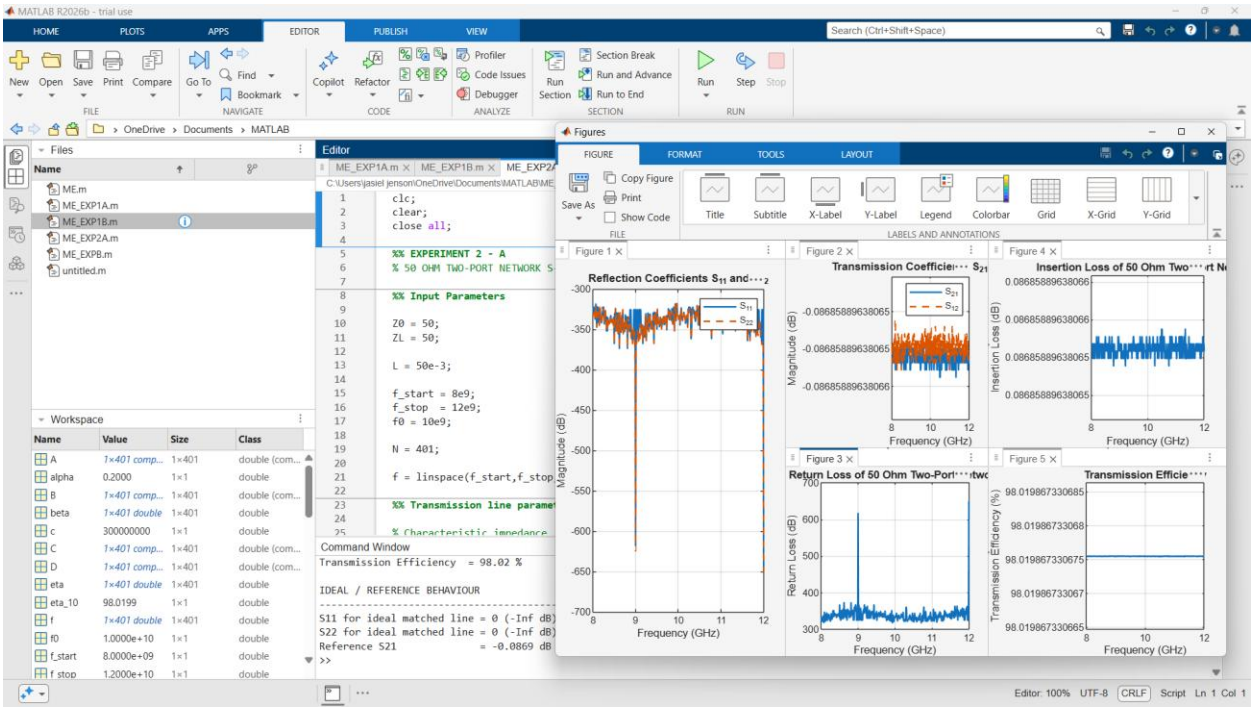
Aim:

To analyze the S-parameters of a 50 Ω microwave transmission line and calculate return loss, insertion loss, and transmission efficiency over 8–12 GHz.

Procedure:

1. Set the characteristic impedance to 50 Ω and line length to 50 mm.
2. Sweep the frequency from 8 GHz to 12 GHz.
3. Calculate and plot \(S_{11}, S_{21}, S_{12}\), and \(S_{22}\).
4. Calculate return loss, insertion loss, and transmission efficiency at 10 GHz.

Output:



Result:

The S-parameters were analyzed over the specified frequency range, and return loss, insertion loss, and transmission efficiency were calculated to evaluate the transmission line's performance.